

Facebook Ad Campaign Analysis

Analysis of real, anonymized Facebook ad campaign data to find which audiences bring sales at the lowest cost.

Data: [Sales Conversion Optimization](#) (Kaggle). 1,143 ads across 3 campaigns, from a 2017-era Facebook campaign, so it reflects that period, not today's ad market.

Live dashboard: open `dashboard.html` (or host it with GitHub Pages).

What's inside

- `analysis.py` - loads the data, validates it (missing values, duplicates, impossible values), calculates metrics, saves a chart
- `dashboard.html` - interactive dashboard with campaign filter, KPI cards and charts
- `KAG_conversion_data.csv` - the raw dataset
- `segment_performance.png` - chart output

Metrics

- **CTR** = clicks / impressions (is the ad catching attention?)
- **CPA (cost per sale)** = spend / approved conversions (what does one sale cost?)
- **Enquiry-to-sale rate** = approved conversions / total enquiries

Key findings

Segment	Cost per sale
Age 30-34	\$30.88
Age 35-39	\$53.68
Age 40-44	\$68.18
Age 45-49	\$99.76
Men	\$41.44
Women	\$69.70

1. Younger audiences convert much cheaper.

Ages 30-34 cost about a third as much per sale as ages 45-49.

2. Older audiences click more but buy less. CTR

risers with age (0.014% to 0.022%), but sales do

not follow. They click, then don't buy.

3. **Men convert better; women click more.** Men cost \$41 per sale vs \$70 for women, though women have the higher CTR.
4. **Best segment: men 30-34 at \$25.56 per sale. Worst: women 45-49 at \$119.94.**
5. **Campaign 1178 took about 95% of the \$58.7k spend** at \$63.83 per sale, versus \$15.81 for campaign 936. Campaign 916 is too small (about \$150) to judge.

Recommendations

- Shift budget from ages 45-49 (especially women) toward ages 30-39.
- For older audiences, test a different offer or landing page. Clicks are not the problem, conversion is.
- Investigate why campaign 1178 costs 4x more per sale than 936 before spending more on it.

Limitations

- The data has no dates, so trends over time cannot be analyzed.
- Only 4 age groups and 2 genders are included.

- Cost per sale is a blended figure. Ads with zero sales are included, which is realistic.

Run it

```
pip install pandas matplotlib  
python analysis.py
```